A Bio-Inspired Minimal Model for Non-Stationary K-Armed Bandits

Krubeal Danieli[1] and Mikkel Elle Lepperød[2]

[1]Center for Integrative Neuroplasticity, FYSCELL, University of Oslo, Norway
[2]Simula Research Laboratory, Oslo, Norway

Abstract

While reinforcement learning algorithms have made significant progress in solving multi-armed bandit problems, they often lack biological plausibility in architecture and dynamics. Here, we propose a bio-inspired neural model based on interacting populations of rate neurons, drawing inspiration from the orbitofrontal cortex and anterior cingulate cortex. Our model reports robust performance across various stochastic bandit problems, matching the effectiveness of standard algorithms such as Thompson Sampling and UCB. Notably, the model exhibits adaptive behavior: employing greedy strategies in low-uncertainty situations while increasing exploratory behavior as uncertainty rises. Through evolutionary optimization, the model's hyperparameters converged to values that align with the principles of synaptic mechanisms, particularly in terms of synapse-dependent neural activity and learning rate adaptation. These findings suggest that biologically-inspired computational architectures can achieve competitive performance while providing insights into neural mechanisms of decision-making under uncertainty.

1 Introduction

The ability to make choices for long-term reward maximization is a fundamental aspect of cognition. The brain has evolved specialized and interconnected regions to implement this behaviour under the constraints of biology.

An established framework for investigating decision-making is reinforcement learning and Bayesian Decision-Theory (BDT), which helped formalizing some possible cognitive strategies in relation to reward [**?**]. In this picture, the decision process possibly relies on past information as well as predictions about the future, whose outputs may affect the eventuality of a positive or negative gain [?]. Important lines of research in this regard revolved around the underlying learning mechanisms, value estimation, and competition among conflicting options. One of the classical learning paradigms that have been suggested is Pavlovian conditioning, where the animal forms direct pairings of external cues and hard-wired responses [1]. A more flexible variant is operant conditioning, in which the response of the animal can be reshaped through repeated experiences and feedback [2]. Another perspective on decision-making is more computational and has the brain abstracted in terms of an agent or controller [?]. Here, proposed approaches may be classified as model based or model free, whose main difference is whether the calculation of the behavior involves an internal model of the environment or only a state-action mapping, with consequences on adaptability and computational efficiency [?]. Concerning their possible brain implementation, the coexistence of a number of controllers has been hypothetized, which may vary in local arrangements, plasticity details, and position within a neuroanatomical hierarchy [3]. Dopaminergic neurons in the ventral striatum of the midbrain have long been associated with model-free algorithms, such as temporal difference learning, given their involvement in action selection and the strong correlation with reward prediction errors [4, 5, 6]. Nevertheless, it has also been speculated that they contributed to model-based dynamics, in particular through the interplay with the orbito-frontal cortex (OFC), a region known for being involved in value representation [7, 8, 9, 10]. Another area that emerged for its role in option evaluation and selection is the anterior cingulate cortex (ACC), particularly during action selection, perceptual disambiguation, and multi-stability situations [11, 12]. Indeed, there is consistent evidence that the interaction between OFC and ACC plays a crucial role in decision-making, integrating emotional and high-level information sourced from the amygdala, and the temporal and hippocampal regions [13, 14, 15, 16].

Well-studied ecological settings for decision making are foraging tasks, such as food search. In these problems, the agent is usually asked to choose between different options to maximize an expected reward. In nature, animals have been shown to exhibit different strategies depending on the context. *Matching behaviour* is a well-known phenomenon in which the animal's decision patterns are proportional to the reward probability of the available options. This behavior is believed to be the result of the trade-off between exploration and exploitation [?]. In fact, this is a well known phenomenon in the reinforcement learning literature, in which an agent is faced with the dilemma of exploring new alternatives, potentially more rewarding, or exploiting known options, despite being possibly sup-optimal.

A popular formalization of these types of task is the *multi-armed bandit* problem (MAB) [?]. This setting is usually described in terms of a slot machine endowed with K distinct arms, also called levers. During a round, the agent selects one of the arms and collects a reward R according to an unknown probability of the specific reward for the chosen arm. The goal is simply to maximize the total reward after a given number of steps, which is achieved by effectively updating a selection policy after each round. This problem has been extensively studied in the context of reinforcement learning and is considered a fundamental building block for more complex tasks [**?**].

The multi-armed bandit problem comes in several variants, with the simplest featuring a stationary reward distribution. An important performance measure in these tasks is *regret*, usually defined as the distance between the selected choice and the theoretically optimal one. Researchers have proposed numerous algorithms to address this problem, each with distinct theoretical guaranties.

Thompson sampling (TS) is a widely adopted approach rooted in Bayesian optimization. It maintains a posterior distribution over action reward probabilities and selects actions by sampling from these distributions. Thompson sampling has demonstrated near-optimal regret bounds in stochastic settings [**?**, **?**]. In contrast, the Upper Confidence Bound (UCB) algorithm uses an optimistic principle for exploration. It maintains an estimate of the reward for each option by a confidence interval. Action selection relies on the upper bound of this interval, encouraging exploration of less-visited options by assigning them higher uncertainty. UCB has been shown to achieve logarithmic regret in stochastic bandits [**?**]. Another effective baseline is the ϵ-Greedy strategy. At each decision step, a random action with probability ϵ and the best known action with probability $1 - \epsilon$ (exploitation) is selected. Although not as theoretically optimal as Thompson Sampling or UCB, ϵ-Greedy is simple to implement and often effective in practice. Extensions such as VDBE adapt ϵ dynamically based on the variance of the value function, providing better control over exploration [**?**, **?**, **?**, **?**].

However, these traditional algorithms, despite their effectiveness, lack biological plausibility – they neither resemble neural circuits nor follow synaptic plasticity dynamics. For example, they do not rely on a network-like architecture with interconnected units, as seen in the brain. Additionally, their action selection process is typically instantaneous, whereas decision making in the brain occurs over time, often requiring the activity of a neural circuit to evolve and stabilize before converging on a final selection. Their learning mechanisms also differ fundamentally. They typically involve explicit updates to statistical parameters (e.g., reward estimates or exploration rates) based on observed outcomes. In contrast, biological learning relies on local plasticity rules, where synaptic changes depend on the activity of connected neurons, modulating how input is integrated and how output signals are generated. Although not the primary driver, these limitations align with a growing interest in machine learning towards bioinspired algorithms, such as neural networks and predictive coding [**?**, **?**], which offers several advantages.

In fact, some of these biological cognitive systems can achieve state-of-the-art performance in various domains, including the challenging *machine-challenging tasks* (MCTs), set of problems that are difficult for machines but relatively easy for humans [**?**, **?**, **?**]. In addition, bioinspired models improve algorithmic interpretability by clarifying the functional relationships between internal components. When applied to tasks with existing experimental data, these models can generate new insights into the brain and suggest new research directions [**?**]. Although other approaches such as Bayesian learning can demonstrate optimal performance and match human data well [?], they are more difficult to relate to neuronal dynamics.

In this work, we aim to enhance the biological plausibility of models used in multi-armed bandit tasks by introducing a novel, minimal decision-making architecture called Neural Selection Agreement model (NSA). This model comprises two interacting rate-based neuronal populations connected by plastic synapses, uses a biologically inspired plasticity rule, and forms decisions based on the agreement of the two populations on the next option.

The model's plasticity mechanism is non-Hebbian and depends on the magnitude of inter-population synaptic weights. This formulation aligns with synapse-type specific plasticity (STSP), a biologically supported mechanism linking learning dynamics to synaptic resource availability, current state, and morphological properties [?, ?, ?, ?]. Learning rates are also adaptive, in line with observations in human experiments [?].

Similar forms of plasticity have been employed in prior work on spiking neural networks and models of synaptic metaplasticity [?, ?]. Despite its simplicity, our model performs comparably with standard algorithms such as Thompson Sampling, ϵ-Greedy and Upper Confidence Bound, while offering a more neurobiologically grounded account of decision-making. Other studies have also proposed solutions to bridge reinforcement learning and neural mechanisms. For example, [?] proposed temporal difference algorithms of the belief state [?] that abstract dopaminergic signaling and medial prefrontal projections. These models highlight the role of hidden-state inference during probabilistic tasks, although they assume fixed reward distributions and do not incorporate explicit synaptic plasticity.

In [?], metaplasticity mechanisms were explored in relation to the probability estimation of binary sequences, uncovering informative patterns of functional synaptic states. However, the environment varied along a single stimulus dimension. Similarly, [?] applied a metaplasticity model to a probabilistic reversal learning task, effectively a stationary two-armed bandit, revealing the emergence of option-specific learning dynamics. A notable exception is [?], which addressed a non-stationary two-arm bandit using a synaptic cascade model [?] equipped with a surprise detection mechanism to track changes in reward probability.

In contrast to previous work, our study addresses more challenging, high-dimensional nonstationary reward environments, including up to 1,000 arms with independently drifting reward probabilities. We specifically focus on stochastic bandit problems with *concept drift*, where reward distributions evolve over time, either gradually or through abrupt changes, thus requiring flexible and adaptive decision-making strategies [?, ?, ?]. Despite its simplicity, the proposed model performs competitively with standard algorithms such as Thompson Sampling, ϵ-Greedy, and Upper Confidence Bound, while offering a more biologically grounded perspective on decision-making mechanisms.

In general, our work aims to bridge adaptive decision-making under uncertainty with principles from computational neuroscience. By proposing a biologically plausible mechanism for choice behavior in dynamic environments, we contribute a framework that may inform both the development of adaptive artificial systems and the interpretation of neural processes underlying flexible behavior.

The remainder of this paper first describes our model design and learning, then presents experimental results and comparative analyses with established algorithms, and lastly discusses the findings' broader implications and potential future directions.

2 Methods

The following section is organized as follows. First, we introduce a formalization of the general problem setting, together with the variants considered in this work. Then we outline the architecture of our model and how it can be mapped to neurobiology. Finally, we describe the learning procedure and showcase its dynamics in a simple example.

2.1 Binomial MAB problem

The standard formulation of the task is structured as a set of K arms (or levers) $\mathcal{A}_K = \{a_1 \dots a_K\}$, with an associated reward probabilities $\mathbf{p} = \{p_1, \dots p_K\}$. At each iteration, the agent pulls an arm and collects a possible reward drawn as a Bernoulli variable $R \sim \mathcal{B}(\{0,1\}, p_k)$. The agent's objective is to maximize the total reward $\sum_t^T R_t$, after a certain number of rounds T, also called the horizon. Importantly, the agent is unaware of the true probability of reward and therefore has to make its decisions following a certain policy, denoted π. In the reinforcement learning literature, the policy is often defined as a distribution over actions, here the arms $\mathcal{A}_K$, given the current state at time t. In the bandit problem, the state can be taken to correspond to the history h_t of past actions and rewards in the period $(0 \dots t]$, and the policy as a function that returns a selected arm $\pi(h_t) = a_t$ [**?**].

Given the inherent stochasticity of the feedbacks from the environment, the policy is affected by the so-called exploration-exploitation trade-off, which here is phrased as the contrast between the option of the arm with the estimated highest expected reward versus the option to explore other arms, so as to gather more information. A common approach is the ϵ-Greedy policy, where the choice to explore is selected with probability ϵ. Moreover, it is often preferable to have more explorative behavior early during the training, with the intent to have a good sample size for the empirical reward distribution, which can be later exploited for maximizing reward.

Another important concept in multi-armed bandit problems is *regret*. Intuitively, it quantifies the loss of reward due to following a certain policy, and it is determined by the difference between the collected reward and the theoretical optimal, obtained by choosing the best arm at each round. Formally, given defined a function $r(\pi)$ that returns the expected reward while following policy π, the regret ρ over an horizon T can be formulated as:

$$\rho = \frac{1}{T}\sum_{t}^{T} p_t^* - r(\pi(h_t)) \tag{1}$$

where p_t^* is the expected reward of the optimal arm at time t, which corresponds to its probability since it is a Bernoulli distribution. The goal of the agent is to minimize the regret and thus maximize the total reward.

2.2 Neural Selection Agreement model (NSA)

The model is constructed as a rate network composed of two neuronal populations, U and V. The first (U) represents the traces of the K available options (that is, the bandits), while the second, V, encodes their values according to the current policy.

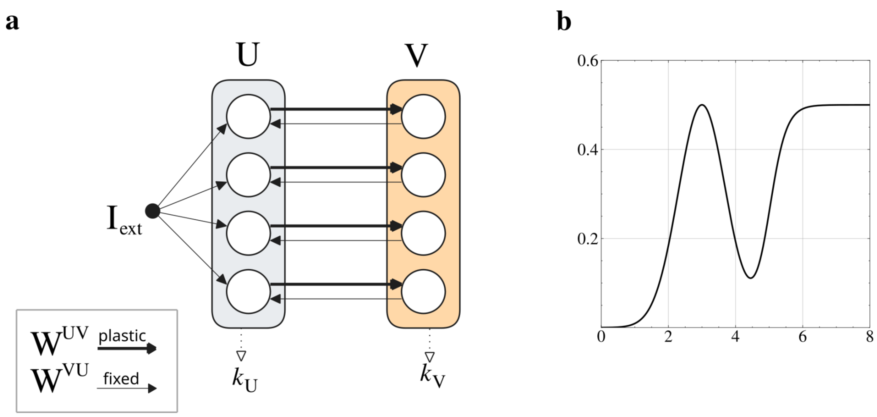

Figure 1: Model architecture - **a**: *the model is composed of a layer U (grey), receiving a feedfoward input I_{ext}, a layer V (orange), and connections $\boldsymbol{W}^{UV}$ and $\boldsymbol{W}^{VU}$. Additionally, two indexes k_U, k_V are extracted from the layers and corresponds to the selection made by the two populations as $k_U = argmax_k\{\boldsymbol{u}\}$, $k_V = argmax_k\{\boldsymbol{v}\}$.* - **b**: *the function used to filter the weights (Φ_v) and the learning rate (Φ_η). It is defined as a weighted sum of a generalized sigmoid and a Gaussian function, both of which have additional parameters controlling their offset and slope steepness (or width). For more details about its implementation see the Appendix 5.2.*

In model, the first layer represents the available options, while the learned connections to the second layer encode their values based on recent reward history. In figure **1a** a visualization of the architecture is presented. A key simplification is the lumping of option representations into single neurons. Although this choice abstracts the more distributed encoding found in actual brain networks, it allows for a more tractable model design [**?**].

More formally, the model is defined by a set of coupled ordinary differential equations (ODE). The first equation describes the evolution of neural activity $\mathbf{u}$ in population U, while the second governs the activity $\mathbf{v}$ in population V, each evolving with its respective time constant τ.

$$\begin{aligned} \tau_u \dot{\mathbf{u}} &= -\mathbf{u} + \mathbf{W}^{VU} \phi_v(\mathbf{v}) + \mathbf{I}_{\text{ext}} \\ \tau_v \dot{\mathbf{v}} &= -\mathbf{v} + \widetilde{\mathbf{W}}^{UV} \phi_u(\mathbf{u}) \end{aligned} \tag{2}$$

The external input $\mathbf{I}_{\text{ext}}$ is a constant input that is used to set the initial conditions of neural activity $\mathbf{u}$. Notably, it does not deliver any practical information, given the unambiguity of the task, and should thus be treated more as an attention-related stimulation to prompt processing [17]. The activation functions ϕ_v, ϕ_u are applied to population v and u respectively, and represent two distinct neural response functions tailored to each population vector. They have been chosen to be a step function with threshold θ_v, θ_u applied to a generalized sigmoid with gain g_v, g_u and offset s_v, s_u.

Importantly, the two layers are not fully connected and the matrices are diagonal. Moreover, the weight matrix $\mathbf{W}^{VU}$ is a diagonal matrix of made of 1s and are fixed, while $\widetilde{\mathbf{W}}^{UV}$ is a function of the actual weights $\Phi_v(\mathbf{W}^{UV})$ and represents the contribution of the active options $\mathbf{u}$ to the value representation $\mathbf{v}$, so it is called *the option value function*. The matrix $\mathbf{W}^{UV}$ is initialized to all zeros and is the only plastic parameter of the model during the task. Concerning the others, they were optimized offline with the goal of maximizing the average total reward over multiple runs and across testing environments. In particular, given the non-differentiability of the model with respect to the fitness function we relied on an evolutionary search using the Covariance-Matrix Adaptation algorithm (CMA-ES) [18].

The function Φ_v is defined to be a combination of a Gaussian and a sigmoid, and its parameters were also optimized offline. **1b** illustrates an example. The motivation behind this choice is offloading to evolution the responsability of converging to a shape useful for solving the task.

The main difference between the two function is the presence (sigmoid) or absence (Gaussian) of a saturation limit on either side of their maximum, with variable offset (threshold) and steepness (gain). Our intent was to explore the effect on performance of the integration and competition of these traits. Furthermore, these features conforms with some characteristics of biological and artificial neurons, such as the firing rate functions being smoothly bounded and receptive fields being defined in a range [19, **?**, **?**, **?**].

Nonetheless, beside these reasons, other basis function could have been chosen with possibly not dissimilar results.

2.2.1 Option selection

The decision-making process within a single round is structured in two distinct phases. Initially, the model receives a constant external input that targets all neurons in the population U equally. During this phase, $\mathbf{I}_{\text{ext}}$ works as an equilibrium value while reciprocal interactions with population V push $\mathbf{u}$ to different values, depending on the current policy encoded in $\widetilde{\mathbf{W}}^{UV}$. Importantly, the weights $\mathbf{W}^{UV}$ are initialized to zero, and thus the input from U to V is uniform. This approach ensures the absence of biases towards any arm by having all weights equal, and corresponds to a completely untrained network. The neural dynamics 2.2 are simulated through a first-order explicit Euler integration method, using a time discretization of 1 (millisecond). After a fixed time duration dur_{pre}s, the second phase begins. Here, the external input is removed and the model is left to evolve autonomously, and since there are no recurrent connections in neither population, the dynamics are entirely driven by their coupling. A selection k is sampled after another fixed amount of time duration dur_{post}s, and is defined according to the following rule:

$$k = \begin{cases} \text{argmax}_k\{\mathbf{v}\} & \textit{if } \text{argmax}_k\{\mathbf{v}\} = \text{argmax}_k\{\mathbf{u}\} \\ \text{random}(K) & \textit{otherwise} \end{cases}$$

The selection rule is simple: if the value representation $\mathbf{v}$ is in agreement with the trace $\mathbf{u}$, then the option with the highest value is selected. Otherwise, a random option is chosen. This rule presents a way to express the exploration-exploitation trade-off through the possible agreement of the two populations, under the influence of current weight values $\widetilde{\mathbf{W}}^{UV}$.
In subsection 2.2.1, the pseudo-code for the algorithm behind the selection process is reported below, which is applied during each round t. The two durations hyperparameters dur_{pre} and dur_{post} are evolved through evolution alongside the others within a range of $(400, 3000ms$. Although this range has been so defined out of empirical motivations and it closely depends on the settings of our model, at least for cases with few options it does align data from experiments measuring reaction times, typically in the order of 300-1500ms [20, 21].
In the computation of the argmax, in case of a tie between options the first result is taken. In fact, empirically no difference has been observed from random sampling, and intuitively in the long term the subset of rewarding options will shrink with learning. According to the values of the policy parameters, the behavior of the model displays periods of exploration followed by steady exploitation, which can be reverted in the case of a change in the environment's reward distribution.

The structure of this option selection process draws inspiration from the interaction between the values estimation, feedback integration, and option representation produced by the orbito-frontal cortex (OFC) and the anterior cingulate cortex (ACC) [9, 14, 16]. Particularly, consensus-like dynamics have been proposed for selecting actions, evaluated in a parallel and distributed fashion, and undergoing competition [22, 23, 24]. Another relevant perspective is

Algorithm 1: Two-phases option selection process

```
Input: External input I_ext, population u, population v, weights W̃^{UV}
Output: Selected action k
Phase 1: external input ;                     // Duration: dur_pre
Define constant I_ext;
Update populations u, v according to 2.2;
Phase 2: autonomous evolution ;               // Duration: dur_post
Remove external input I_ext;
Update populations u, v according to 2.2;
Selection process:;
k_u ← argmax_k{u};
k_v ← argmax_k{v};
if k_u = k_v  and  k_u > 0 then
 |  k ← k_v ;                                 // Exploitation
else
 |  k ← random(K) ;                           // Exploration
end
return k
```

that of bump attractor models for perceptual decision-making, usually defined as recurrent networks trained to drift within their activity landscape to solve multi-choice tasks [25, 26, 27].

2.3 Learning

Given a selected option k, the environment (set of bandits) samples and returns a reward $R = 1$ with probability p_k. Then, the weights $\mathbf{W}^{UV}$ for the neuron corresponding to option k are updated according to the following plasticity rule.

$$\Delta \mathbf{W}_k^{UV} = \tilde{\eta}_k \left(R \cdot w^+ - \mathbf{W}_k^{UV} \right) \tag{3}$$

where w^+ is a constant maximum synaptic weight, while $\tilde{\eta}_k$ is the learning rate for the option k determined by a function Φ_η of the current weights $\mathbf{W}_k^{UV}$, known as the *learning rate function.*

The shape of Φ_η is again a Gaussian-sigmoid but with different parameters, giving evolution the opportunity to combine the two characteristic traits of smooth saturation and bell-shaped tuning. In terms of neural dynamics, the advantage is to introduce adapation in the learning of the weights, implemented as a more considerate scaling of the update magnitude. Computationally, this allows taking into account different learning phases for a given neuron, which represents an option, and approach similar to adaptive optimizers employed in training deep networks [28]. Further, proportional updates and synaptic scaling have been observed in biological synapses, taking part plasticity mechanisms, like shaping the synaptic update kernel, or homeostatic constraints, for instance limits in the speed of chemical exchange in the synaptic cleft [?, ?, ?, ?].

3 Experiments

The NSA model was tested in a series of benchmark environments, each with a different number of arms and reward distributions. Performance has been compared with the following algorithms: Random Baseline, Upper-Confidence Bound (UCB), Thompson Sampling, and Epsilon-Greedy. The first of two of these were implemented free from any pre-defined hyper-parameters, such that they automatically balanced exploitation-exploration with experience. However, $\epsilon-$Greedy necessitated the tuning of its hyper-parameter ϵ, for which we defined a calculation based on the number of options of the form $0.4 * \log_2(0.6 * K)$, whose specific values were fitted empirically.

3.1 Game variants

Our goal is to investigate the performance of the agent in a non-stationary environment, meaning that its underlying distribution changes over time [1] We choose this setting as it resembles an environment in which an animal has to forage in an environment with food (reward) distributed over a set of fixed locations, but whose occurrence probability can change over time. A *round* -or horizon - is defined as an action-reward event; instead, a *trial* is a block of rounds. For testing, four slightly different MAB variants were used, obtained by introducing different types of non-stationarity: piecewise constant, uniformly changing, sinusoidally changing, and sinusoidally changing with piecewise constant arms. The reason for these choices is to test the performance of the model at different speeds and uniformities in the distribution changes. Figure 2 visually illustrates their specificities.

Piecewise stationary distribution [MAB-P]
Within a trial, the reward distribution is stationary and is drawn from a normal $\mathbf{p} = \mathcal{N}(0.5, 0.2)^K$, clipped in $(0, 1)$. At the end of each trial i, a new distribution $\mathbf{p}_i \to \mathbf{p}_{i+1}$ is drawn [?].

Piecewise stationary distribution with drift [MAB-D]
At first, the reward distribution $\mathbf{p}$ is sampled from a normal $\mathbf{p} = \mathcal{N}(0.5, 0.2)^K$. Then, it changes gradually over the rounds, tracked over time t, such that its values tend towards a target distribution $\mathbf{q}_i$ as $\tau_p \dot{\mathbf{p}}_t = \mathbf{q}_i - \mathbf{p}_t$. Here, $\dot{\mathbf{p}}$ is the time derivative of the distribution, and τ_p is its time constant. Once the distance is below a threshold δ as $|\mathbf{q}_i - \mathbf{p}_t| < \delta$, the target distribution is changed to a

[1]Since the arm probabilities are not normalized to 1, it is technically improper to call them *probability distributions*; we will therefore refer to either *probability* or *distribution* separately at any given time to avoid confusion.

new one $\mathbf{q}_i \rightarrow \mathbf{q}_{i+1}$. In this variant, there are no proper trials, but the target distribution keeps changing until a maximum number of rounds is reached.

Sinusoidal distribution shift [MAB-sin]
The reward distribution changes over rounds, with the probability of each arm following a sine wave with a specific frequency f_k, phase λ_k and amplitude 1. At any given time t, the distribution is $\mathbf{p}_t = \{\sin(2\pi f_k t + \lambda_k)$ for $k = 1 \ldots K\}$.

Partial sinusoidal distribution shift [MAB-sinP]
Identical to the sinusoidal distribution shift, half of the arms change sinusoidally, while the other half is always kept at a constant value. The distribution is not normalized.

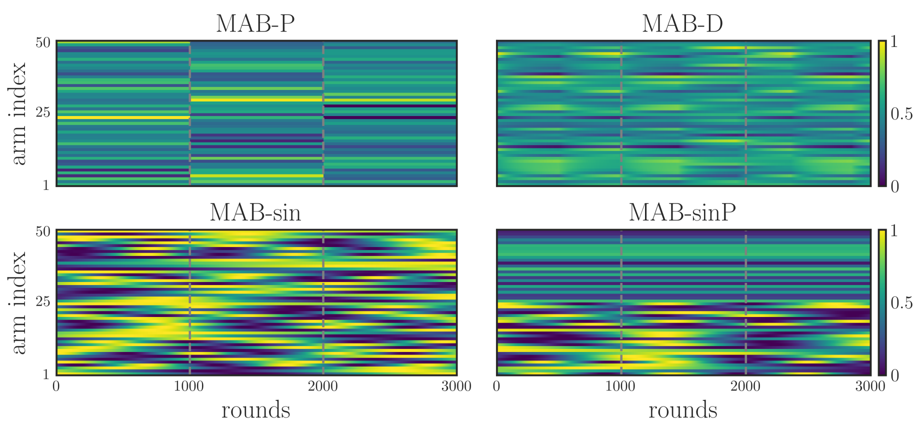

Figure 2: Reward distribution for the four MAB variants - *The reward distribution for each variant is illustrated: piecewise stationary distribution (MAB-P), piecewise stationary distribution with drift (MAB-D), sinusoidal distribution shift (MAB-sin), partial sinusoidal distribution shift (KAB-sinP). The arms are organized in rows; the grey dashed lines demarcate trials (3), which are block of rounds represented by columns, and here only reporting the arm reward proability.*

3.2 Evolutionary search

The optimization of the hyper-parameters was performed using the Covariance Matrix Adaptation evolutionary strategy algorithm (CMA-ES) [**?**]. The search was carried out with a population of 256 individuals for 80 generations. Each individual was endowed with a genome, corresponding to a vector of 22 parameters of the model. The fitness function of the evolution was defined as the average reward obtained by an individual over 3 different non-stationary bandit environments, each for $K = \{40, 200\}$, and all averaged over 2 iterations. The results are summarized below in figure 3.

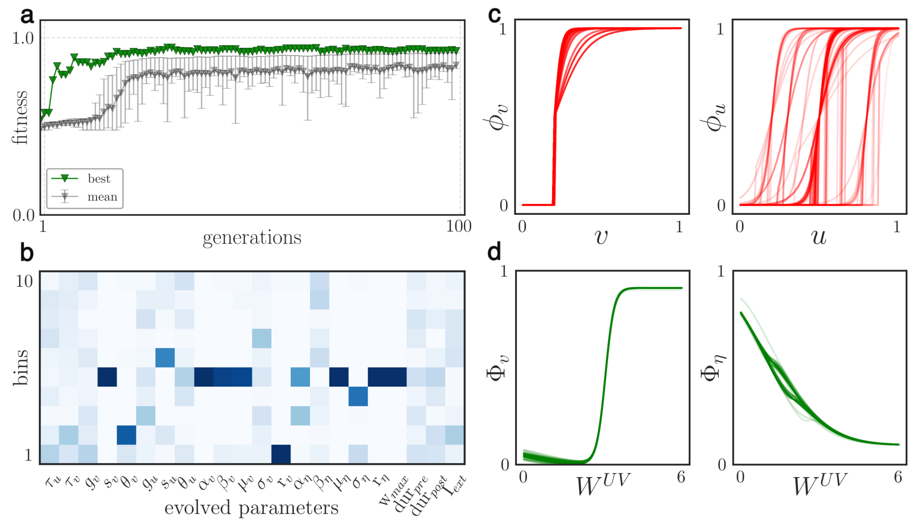

Figure 3: EVOLUTION RESULTS - **a**: *top fitness, mean and standard deviation (as 16-84 percentile) of the population over generations.* - **b**: *heatmap of the evolved parameters (rows) as histogram bins (y-axis) calculated from the 50 percentile of the population of the last generation; higher density is in dark blue* - **c-d**: *neural response functions [c] and Gaussian-sigmoid [d] of the top-half of the population, the color intensity is proportional to the fitness.*

The evolution progress, plot **3a**, showed a steady improvement over the generations, with the fitness hitting a plateau once the total reward could not get any closer to the optimum ≈ 1 due to randomness and the finite horizon.

The evolved parameters (constituting a *genome*) are displayed as a frequency heatmap in **3b**, which was obtained from individuals whose fitness was equal to or above the median. The parameters most strongly influencing the results were those directly associated with neuronal activity and learning. For each, the sensitivity of each value with respect to performance is inversely proportional to the variance its distribution over the population (related to the heatmap color intensity in **3b**). The intuition is that the higher the variability the smaller is the number of top-scoring genomes sharing a similar value.

Regarding the neural response functions shown in **3c**, both populations evolved to have a similar shape, a sharp sigmoid with a clear threshold, with population U having a more variable distribution. The form is characterized by not allowing for a fine-grained linear reponse but rather a high-pass filter, with activity occurring only after strong excitation. This firing behaviour is reminiscent of coincidence detector neurons, which are sometimes referred to as class III neurons with respect to their f-I curve [**?**].

The two Gaussian-sigmoid functions Φ_v, Φ_η, which are, respectively, the option value and learning rate functions, are instead shown in **3d**. On the one hand, the learning rate function Φ_η is characterized by a decreasing curve, associated with the dominance of the right-side of the Gaussian, and a completely ignored sigmoid component. A possible interpretation for this shape is to ensure a high learning speed when the value options are more uncertain (weak weights), and low otherwise; thus preventing overshooting and oscillations in the weight updates.

On the other hand, the option value function Φ_v instead follows a steep sigmoid curve. This is consistent with the idea that the input of population U to population V is weighted maximally for high option values (strong synapses), whereas for weaker estimates the contributions are low or close to zero, allowing for more exploration. Interestingly, a common feature seemed to be a slight concavity after zero, a slim influence of the Gaussian component, which might be interpreted as a sort of test for newly formed synapses. However, the size of this effect is not large.

3.3 Environment variants and number of arms

The NSA model has been tested and compared with the other algorithms: Thompson Sampling (TS), UCB, and ϵ-Greedy. The benchmark were the four different variants of the MAB problem listed above 3.1, with a variable number of arms ranging from 5 to 1000. The results are reported in table 1. Overall, our NSA model displays solid performance in all environments, most of the time being equally good or better than the other algorithms. Interestingly, a large number of arms (K) did not present a significant challenge, as the model was effectively adapted to the different environments and reward distributions. However, this is in part due to the randomness in the assignment of arm probabilities, and the statistics of the quantity of high-reward arms as their number increases. Nonetheless, given the non-stationarity of the reward distribution it is still a non trivial task to re-calibrate to new distributions.

3.4 Analysis of dynamics and robustness

3.4.1 Entropy analysis

For a better understanding of the qualitative differences between the models, we analyzed the progress over the rounds by tracking the selected arms in a simple piecewise stationary distribution environment with $K = 200$. The simulation was run for 2 trials with 2000 rounds each and then averaged over 40 iterations. In order to quantify the variability of the decision policy at a given time and highlight the particularity of each decision-making behavior, we calculated the entropy of the probability distribution p of the chosen arms, calculated over a window of 20 rounds, as $H = -\sum_i^K p_i \log(p_i)$. The unit of entropy is in nats, and it ranges from 0 (no uncertainty) to $\log_e(K)$ (maximum uncertainty). In Figure 4-**a**, the raster plot of the selected arms is plotted for each model together with its level of entropy. The distribution of the probability of reward on the arms has an average of $H = 2.02$.

As expected, the shape of the entropy curve expresses the inherent strategy

	K	5	10	50	100	200	1000
MAB-P	TS	**0.03**(**8**)	**0.02**(**6**)	**0.02**(**7**)	**0.04**(**5**)	**0.02**(**3**)	0.16(2)
	ϵ-Greedy	0.05(14)	0.07(5)	0.08(7)	0.15(6)	0.08(2)	0.10(4)
	UCB	0.05(15)	0.05(8)	0.19(6)	0.33(3)	0.39(2)	0.54(3)
	NSA	*0.08(13)*	*0.07(11)*	*0.07(14)*	*0.07(8)*	*0.09(9)*	**0.07**(**8**)
MAB-D	TS	**0.03**(**6**)	**0.08**(**13**)	0.16(6)	**0.19**(**3**)	0.28(7)	0.34(3)
	ϵ-Greedy	0.04(7)	0.14(13)	0.22(5)	0.19(8)	0.26(7)	0.16(4)
	UCB	0.05(6)	0.09(13)	0.21(3)	0.36(4)	0.40(3)	0.49(2)
	NSA	*0.13(10)*	*0.15(16)*	**0.05**(**6**)	*0.21(5)*	**0.26**(**7**)	**0.12**(**7**)
MAB-sin	TS	0.21(22)	0.22(16)	0.07(5)	0.10(5)	**0.06**(**4**)	0.21(5)
	ϵ-Greedy	0.21(21)	0.18(10)	0.12(5)	0.12(6)	0.10(4)	0.10(1)
	UCB	0.03(4)	0.05(3)	0.17(4)	0.23(1)	0.33(3)	0.49(3)
	NSA	**0.00**(**3**)	**0.02**(**4**)	**0.05**(**3**)	**0.06**(**4**)	*0.08(1)*	**0.05**(**4**)
MAB-sin**P**	TS	0.19(21)	0.43(19)	0.17(10)	0.11(6)	0.09(6)	0.19(6)
	ϵ-Greedy	0.24(26)	0.43(10)	0.24(10)	0.14(5)	0.15(6)	0.14(2)
	UCB	0.00(6)	0.26(17)	0.18(6)	0.29(3)	0.34(1)	0.52(3)
	NSA	**0.00**(**9**)	**0.23**(**17**)	**0.14**(**7**)	**0.08**(**8**)	**0.06**(**4**)	**0.09**(**7**)

Table 1: TABLE OF PERFORMANCE — *From top to bottom: results for MAB-P, MAB-D, MAB-*sin*, and MAB-*sin*P, for different numbers K of arms. Each cell shows average regret and standard deviation (in parentheses), computed over 2 trials of 2000 rounds, averaged over 5 simulations.*

adopted by each model. In particular, the UCB algorithm showed the highest variability, marked by persistent exploratory behavior throughout the trials despite converging to reward options. Thompson Sampling was able to reach most solutions, although it had difficulty adapting to new reward distributions that led to high entropy levels. ϵ-Greedy also showed a good performance quite reliably, with the greedy strategy ensuring low entropy for most rounds. Similar behaviour was observed for NSA, which was able to reach the optimal policy and maintain it over time, with entropy peaking mostly at the beginning of the trials and being, on average, the lowest among all models. Indeed, the dynamics of NSA makes it particularly suited for the task of non-stationary MAB, as it is able to quickly adapt to new reward distributions and firmly maintain a greedy policy.

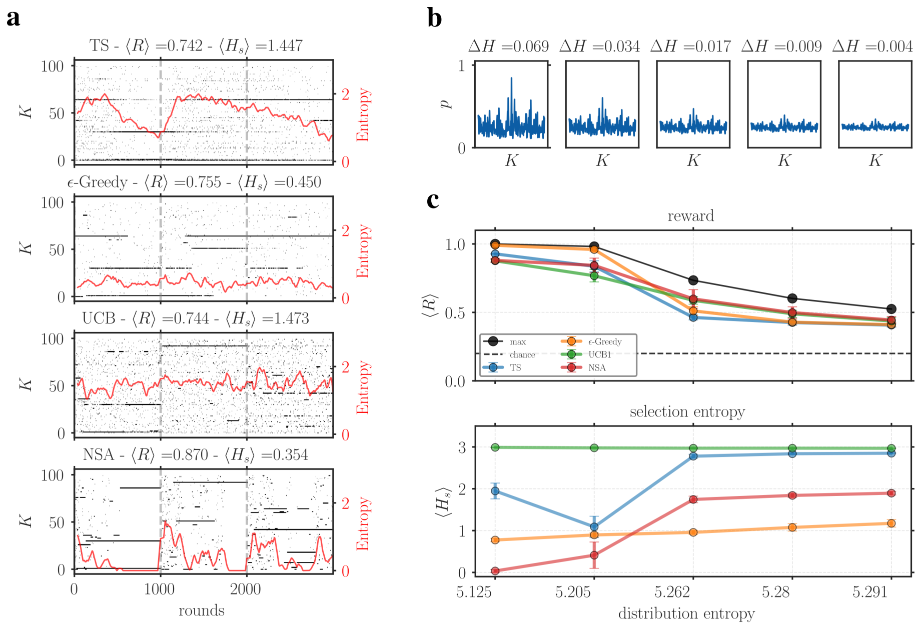

Figure 4: Decision-making dynamics for different models - **a**: *Each plot display the results from one model. The raster plots (black dots) show the arms selected at each round. The red lines represent the entropy level, calculated from the distribution of selections over the preceeding 20 rounds, smoothed with a 30-steps moving average. In the plot titles, the total reward and average entropy (H) over all trials are also reported.* - **b**: *series of distributions of arm probabilities ($K = 200$) with increasing levels of entropy. Subplot titles report the difference ΔH with respect to a reference uniform distribution with same mean, representing the limit of the distribution series with maximal entropy.* - **c**: *the top sub-plot displays the average reward $\langle R \rangle$ for one trial obtained by each model for increasing levels of distribution entropy in the reward distribution. The bottom sub-plot shows the average entropy of the selections $\langle H_s \rangle$ for the same trial. The simulation used a stationary piecewise environment with $K = 200$, 2000 rounds, and averaged over 40 repetitions.*

Then, we sought to investigate the robustness of NSA by targeting the capacity to endure increasing levels of entropy in the reward distribution, defined as $-\sum_i^N p_i \log(p_i)$ and calculated in nats. Figure **4b** showcases an example of arm distributions with increasing entropy, highlighting the blending, in terms of reward probability, of the best arm with the others, making the outlook more homogeneous and uniform. For more details on the implementation, see the Appendix 5.5. The investigation used a simulation carried out in a piecewise stationary environment with $K = 200$, results are plotted in Figure **4c**.

The analysis tracked performance, as cumulative reward, as well as selection entropy H_s, defined as the entropy of the distribution of selected options within a sliding window of 20 rounds.

In the top sub-plot, the average reward obtained by each model is shown against the entropy of the reward distribution.

The results reported how all models are capable of robust performance in the first trial even in the presence of high uncertainty. In the second trial, ϵ-Greedy and Thompson Sampling suffered the increasing difficulty of switching arms, probably due to their conservative approaches. However, this challenge affected UCB and NSA only with higher entropy levels, recognizing their adaptability.

Another perspective on this analysis was given by the bottom plot, which showed the average entropy over the trials. Overall, there was an unsuprising trend of increasing selection entropy with the entropy of the reward distribution. However, striking is the exception of Epsilon-Greedy, which still maintained a constant level throughout. UCB displayed the highest average values, while Thompson Sampling followed with some delay. On the other hand, NSA displays a more abrupt change, going from a state of very low to high variability, a sign of solid exploratory behavior.

4 Discussion

The ability to make decisions under uncertainty is a fundamental aspect of cognition. A well-established framework for studying this capacity is the multi-armed bandit problem (MAB), which has been widely explored and extended across multiple domains [**?**, **?**].

In behavioral experiments, humans demonstrate remarkable adaptability in such settings, integrating environmental uncertainty, generalizing across trials, and dynamically adjusting their learning rates. These behaviors reflect a diversity of cognitive strategies [**?**]. Although Bayesian approaches often capture human behavior well [**?**, **?**, **?**], they are challenging to map directly onto biologically realistic neural dynamics. Despite the existence of many algorithms with strong theoretical guarantees, most lack biological plausibility, particularly in their architectural assumptions and learning and choice mechanisms.

In this work, our aim was to design a minimal, biologically inspired architecture able to solve non-stationary MAB tasks. Specifically, we proposed a simple architecture composed of two interacting and plastic populations of rate-based neurons and producing choices through agreement, called Neural Selection Agreement model (NSA). We evaluated it on four variants of the MAB problem, each differing in how reward probabilities evolved over time and in a wide range of arm counts, from a few options to over a thousand. For comparison, we also tested three standard algorithms.

Our results show the model's ability to adapt to changing reward distributions and quickly recover performance over time. It reliably tracked reward-optimal options and sustained effective decision policies, matching the performance of established methods such as Thompson Sampling, ϵ-Greedy, and Upper Confidence Bound (UCB).

To better understand the behavior of the system, we analyzed its responses at varying levels of reward distribution entropy. In low-uncertainty settings, NSA quickly identified the rewarding option and adopted a greedy policy, similar to Thompson Sampling. In contrast, UCB maintained a higher degree of exploration. As uncertainty increased, the model exhibited a higher option entropy in its decisions, transitioning to a more exploratory strategy similar to UCB. Although this change modestly affected the NSA's ability to switch arms in highly volatile environments, overall performance remained robust.

The strengths of the our model can be traced in both the architecture and the learning paradigm, whose hyperparameters were optimized through an evolutionary process. Interestingly, the values found converged to solutions that can be mapped to plausible synaptic mechanisms. On the one hand, neural dynamics, which rely on plastic connections and a consensus-like selection process. Particularly important was the choice of modulating the afferent connections to the value population V according to a nonlinear function dependent on the synaptic weight itself. In so doing, it was possible to implicitly evolve an effective option value policy for the trade-off between exploration and exploitation.

The neural response functions that emerged were characterized by a steep sigmoidal shape, which can be related to the saturation of the neural response once a certain threshold is crossed, a feature observed in the biological network as class III neurons, in addition to being a common choice for artificial ones [?, ?, ?].

On the other hand, learning was structured as a nonassociative plasticity rule based on the reward. Similarly to before, a non-linear function of synaptic weights played a critical role, specifically in defining the synapse-specific learning rate [?]. Furthermore, the shape evolved of the learning rate function was inversely proportional to the synaptic weight, which can be related to the availability of resources in the synapse and its state, including size [?, ?].

An additional consideration is the inspiration from the functional role of the orbitofrontal cortex (OFC) and anterior cingulate cortex (ACC), two important pre-frontal regions known to be involved in decision-making processes [?, ?].

In our work, we have explored ways in which an option value can be formed according to recent reward history and connection weights, updating the option representations. The OFC is known to represent different options, updating their values based on history and rewards [?, ?, ?].

Next, the NSA model relies on some inductive biases, the shape of the Gaussian-sigmoid function, affecting the neural activity and the weight-dependent learning rate. These biases may be considered to implicitly encode a policy that dictates how the two populations interact, how new information should be incorporated into the update of the weight value, and what option to choose next. In fact, ACC has been associated with the evaluation of actions and the regulation of the balance between exploration and exploitation [?, ?].

Additionally, the generation of an option selection results from a temporal interaction of the activity of the two neural population, before converging to a choices. In a similar direction, it has been observed that the OFC transiently visits chosen and unchosen options before committing [?]. Further, the dynamic interaction between the ACC and OFC has been linked to transient pre-stimulus activations, which bias decisions toward the most valuable option [?, ?, ?].

Despite the promising results, there are some limitations to the model. First, we considered the great level of abstraction in the neuronal details, as we considered simple point neurons with synapses modeled with relatively elementary functions, lacking the anatomical complexity of actual dendrites. In particular, the NSA does not account for the presence of noise in neural dynamics, which is a well-known feature of biological neurons [?]. Furthermore, the functional association with the pre-frontal cortical region is only moderate, although present. On the computational side, since our interest lied in the biological plausibility and evolution of adaptive meta-learning solutions, we used only a few well established and relatively simple algorithms as a reference and did not take into account more advanced variants such as VDBE [?, ?], *f-Discounted-Sliding-Window Thompson Sampling* (*f-dsw TS*) [?], and variants of ϵ-Greedy [?].

Future work could involve comparison with more sophisticated algorithms, the introduction of a larger architecture, and more realistic neural dynamics, such as spiking neurons [?].

Acknowledgements & Statements
The authors declare no competing interests.
The code is publicly available and can be found at `https://github.com/iKiru-hub/minBandit.git`.
This research was funded by the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement N$^{\text{o}}$ 945371 and the University of Oslo.
The research presented in this paper has benefited from the Experimental Infrastructure for Exploration of Exascale Computing (eX3), which is financially supported by the Research Council of Norway under contract 270053.
Lastly, special thanks to Kosio Beshkov and Marianne Fyhn for inputs and feedback.

References

[1] Chris T. Allen and Thomas J. Madden. A closer look at classical conditioning. 12(3):301–315.

[2] Gianluca Baldassarre and Domenico Parisi. Classical and instrumental conditioning: From laboratory phenomena to integrated mechanisms for adaptation.

[3] Dominik R. Bach and Peter Dayan. Algorithms for survival: a comparative perspective on emotions. 18(5):311–319. Publisher: Nature Publishing Group.

[4] Roland E. Suri. TD models of reward predictive responses in dopamine neurons. 15(4):523–533.

[5] Yael Niv, Michael O. Duff, and Peter Dayan. Dopamine, uncertainty and TD learning. 1(1):6.

[6] Wolfram Schultz. Dopamine reward prediction error coding. 18(1):23–32. Publisher: Taylor & Francis.

[7] Michael J. Frank and Eric D. Claus. Anatomy of a decision: Striato-orbitofrontal interactions in reinforcement learning, decision making, and reversal. 113(2):300–326.

[8] Teghpal Singh, Michael A. McDannald, Richard Z. Haney, Domenic H. Cerri, and Geoffrey Schoenbaum. Nucleus accumbens core and shell are necessary for reinforcer devaluation effects on pavlovian conditioned responding. 4:126.

[9] Steven W. Kennerley and Mark E. Walton. Decision making and reward in frontal cortex. 125(3):297–317.

[10] Michael A. McDannald, Yuji K. Takahashi, Nina Lopatina, Brad W. Pietras, Josh L. Jones, and Geoffrey Schoenbaum. Model-based learning and the contribution of the orbitofrontal cortex to the model-free world. 35(7):991–996. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1460-9568.2011.07982.x.

[11] Mark E. Walton, Paula L. Croxson, Timothy E. J. Behrens, Steven W. Kennerley, and Matthew F. S. Rushworth. ADAPTIVE DECISION MAKING AND VALUE IN THE ANTERIOR CINGULATE CORTEX. 36:T142–T154.

[12] Shervin Safavi and Peter Dayan. Multistability, perceptual value, and internal foraging. 110(19):3076–3090. Publisher: Elsevier.

[13] Michael H. Rosenbloom, Jeremy D. Schmahmann, and Bruce H. Price. The functional neuroanatomy of decision-making. 24(3):266–277. Publisher: American Psychiatric Publishing.

[14] Chung-Hay Luk and Jonathan D. Wallis. Choice coding in frontal cortex during stimulus-guided or action-guided decision-making. 33(5):1864–1871. Publisher: Society for Neuroscience Section: Articles.

[15] Zuzanna Z. Balewski, Thomas W. Elston, Eric B. Knudsen, and Joni D. Wallis. Value dynamics affect choice preparation during decision-making. 26(9):1575–1583.

[16] Edmund T. Rolls. Emotion, motivation, decision-making, the orbitofrontal cortex, anterior cingulate cortex, and the amygdala. 228(5):1201–1257.

[17] Steven Yantis. The neural basis of selective attention. 17(2):86–90.

[18] Christian Igel, Nikolaus Hansen, and Stefan Roth. Covariance matrix adaptation for multi-objective optimization. 15(1):1–28.

[19] Daniel A. Butts and Mark S. Goldman. Tuning curves, neuronal variability, and sensory coding. 4(4):e92. Publisher: Public Library of Science.

[20] Anne K. Churchland, Roozbeh Kiani, and Michael N. Shadlen. Decision-making with multiple alternatives. 11(6):693–702.

[21] Jan Drugowitsch, Gregory C DeAngelis, Eliana M Klier, Dora E Angelaki, and Alexandre Pouget. Optimal multisensory decision-making in a reaction-time task. 3:e03005.

[22] Paul Cisek and John F. Kalaska. Neural mechanisms for interacting with a world full of action choices. 33(1):269–298.

[23] Paul Cisek. Making decisions through a distributed consensus. 22(6):927–936.

[24] Shinsuke Suzuki, Ryo Adachi, Simon Dunne, Peter Bossaerts, and John P. O'Doherty. Neural mechanisms underlying human consensus decision-making. 86(2):591–602.

[25] Hongzhi You and Da-Hui Wang. Dynamics of multiple-choice decision making. 25(8):2108–2145.

[26] Sam Carroll, Krešimir Josić, and Zachary P. Kilpatrick. Encoding certainty in bump attractors. 37(1):29–48.

[27] Jose M. Esnaola-Acebes, Alex Roxin, and Klaus Wimmer. Bump attractor dynamics underlying stimulus integration in perceptual estimation tasks.

[28] XiaoHao Wen and MengChu Zhou. Evolution and role of optimizers in training deep learning models. 11(10):2039–2042.

5 Appendix

5.1 Neural response function

The activation functions applied to the two neuronal population are defined as a step-function composed with a generalized sigmoid as follows:

$$f(x; g, o, \theta) = \begin{cases} \left[1 + e^{-g(x-o)}\right]^{-1} & \text{if } \left[1 + e^{-g(x-o)}\right]^{-1} > \theta \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Where:

- x is the neuron pre-activation value
- g is the gain
- o is the offset
- θ is the threshold

Each population has its own set of parameters, which are optimized through evolutionary search.

5.2 Gaussian-sigmoid function

The function Φ. is defined by combining a generalized version of the sigmoid, namely with a gain $\beta \neq 1$ and offset $\alpha \neq 0$, and a Gaussian with mean μ and variance σ^2. Their contributions are weighted by as r and $1 - r$ ($r \in (0, 1)$) respectively.

$$\Phi_v(x) = r \left(1 + \exp^{-\beta(x-\alpha)}\right)^{-1} + (1 - r) \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The motivation behind this choice is to express a function that possesses a bounded region (depending on μ, σ) at a high/low peak (depeding on the value of γ_2), and a continuous transition to a constant value (depending on the steepness of the sigmoid β, shift α, and intensity γ_1).

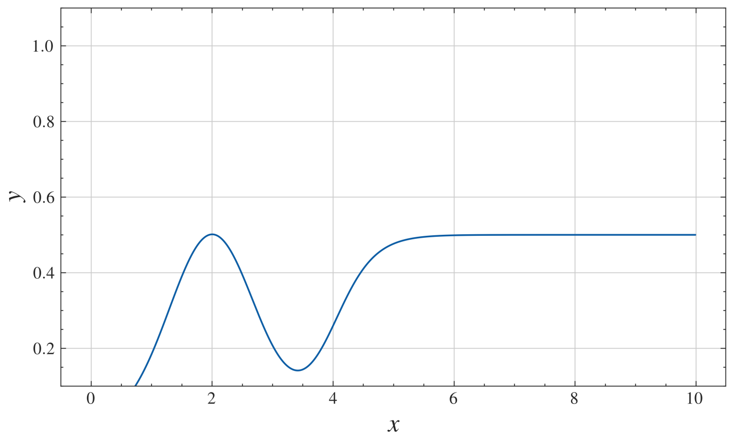

Figure 5: Activation function Φ_v - *Parameters* $\beta = 10$, $\alpha = 1$, $\mu = 1$, $\sigma = 1$, *and* $r = 0.5$.

5.3 Literature models

TODO

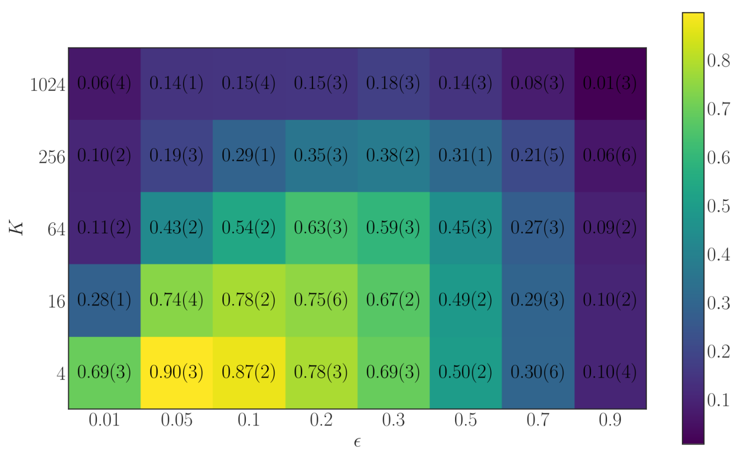

Figure 6: Sensitivity analysis of Epsilon-Greedy - *performance of the algorithm on a piecewise stationary MAB for varying number of arms and* ϵ *parameter*

5.4 Evolutionary search

The optimization was carried out over several parameters concerning the model architecture, its dynamics, as well as three hyperparmeters of the selection process. The following is the entire set of evolved valued, with in parenthesis the range in which they were allowed to be sampled from. The specific choice of range has been chosen by trial and error, such that the ending distribution would not cluster over either the positive or negative (if allowed) boundary, as shown in 7.

Network parameters

- $\tau_u \in (1, 300)$: time constant of population U
- $\tau_v \in (1, 300)$: time constant of population V
- $g_u \in (1, 50)$: gain of the neural response function of population U
- $g_v \in (1, 50)$: gain of the neural response function of population V
- $s_u \in (0.1, 5)$: offset of the neural response function of population U
- $s_v \in (0.1, 5)$: offset of the neural response function of population V
- $\theta_u \in (-0.5, 0.9)$: threshold of the neural response function of population U
- $\theta_v \in (-0.5, 0.9)$: threshold of the neural response function of population V
- $W^+ \in (2, 5)$: maximal weight value for the weights $\mathbf{W}^{UV}$

Option value function parameters

- $\beta_v \in (0.1, 10)$: steepness of the sigmoid
- $\alpha_v \in (-5, 5)$: shift of the sigmoid
- $\mu_v \in (-5, 5)$: mean of the Gaussian
- $\sigma_v \in (0.01, 10)$: variance of the Gaussian
- $r_v \in (-0.5, 1.5)$: weight of the sigmoid

Learning rate function parameters

- $\beta_\eta \in (0.1, 10)$: steepness of the sigmoid
- $\alpha_\eta \in (-5, 5)$: shift of the sigmoid
- $\mu_\eta \in (-5, 5)$: mean of the Gaussian
- $\sigma_\eta \in (0.1, 10)$: variance of the Gaussian
- $r_\eta \in (-0.5, 1.5)$: weight of the sigmoid

Option selection process hyperparameters

- $\mathrm{dur}_{\mathrm{pre}} \in (400, 3000)$: duration of the first phase

- $\mathrm{dur}_{\mathrm{post}} \in (400, 3000)$: duration of the second phase
- $I_{\mathrm{ext}} \in (0.1, 1)$: magnitude of the external input

Each individual has been evaluated over two environments the following settings:

- MAB-0: average reward distribution entropy $\langle H \rangle = 2.05$
- KAB-sinP: average reward distribution entropy $\langle H \rangle = 2.1$, given K arm frequencies f_k as an equally spaced set $\{0.1 \dots i \dots 0.4\}$, phases λ_k drawn from an uniform $\sim \mathcal{U}(0, 2\pi)$, and half of the arms have been set to constant values drawn from another uniform $\sim \mathcal{U}(0.1, 0.7)$; the final reward distribution was not normalized.

The number of arms was $K = 40$ and 200, and lasted for 2 trials with 2000 rounds each. The final fitenss was the average over 2 iterations.

The optimization has been implemented in Python using the `DEAP` library, and the algorithm used was the `CMA-ES` algorithm. The optimization involved 40 generations with a population size of 256 individuals. The mutation rate was set to 0.5 with a sigma of 0.8, the cross-over rate was set to 0.4. The run were carried out on a 256-core AMD EPYC 7763 with 2TB of RAM.

5.4.1 Genome distribution

Following the evolutionary search, it is taken the distribution of parameters over the top-scoring half of the population, corresponding to 128 individuals.

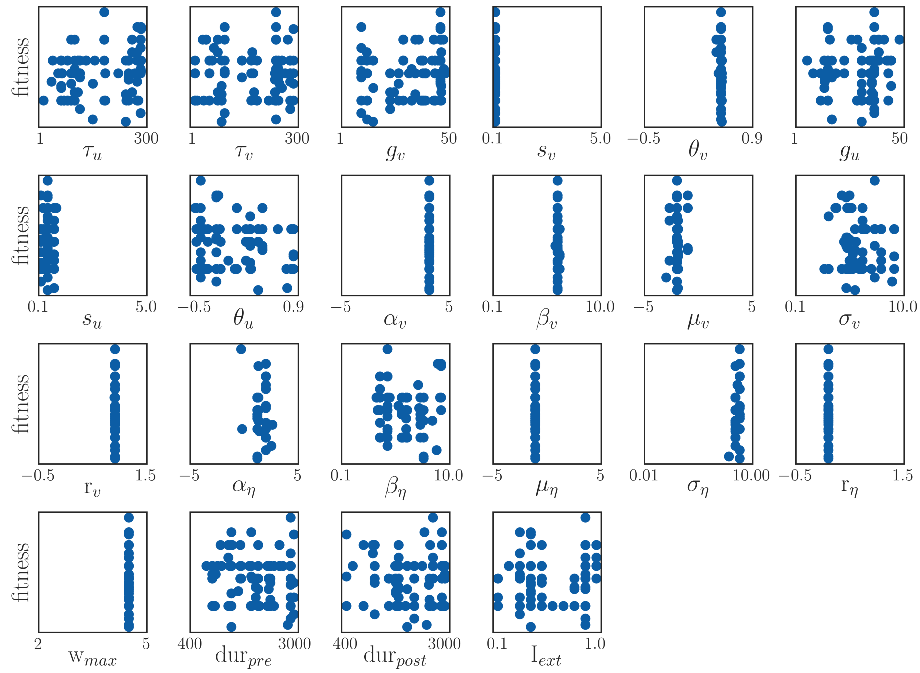

Figure 7: Genome distribution - *each parameter is plotted against the fitness score.*

Figure 7 reveals those parameters shared among the models with the highest fitness score, and those that are more variable. The most stable parameters are, predicatbly, those involved in directly shaping the neural activity (neural activation functions) and learning policy (Gaussian sigmoid).

For what concerns the option selection algorithm 2.2.1, the values for the duration of the two phases resulted to be of not modest importance, with the distribution being weakly skewed towards longer times. As shown in 8, also the fitness is seemingly not associated with particular values for either phase.
The effect of different durations for the first and second phases on performance are limited, as shown in Figure **??**.

5.5 Reward distribution entropy

The calculation of a set of N reward probability distribution $\mathbf{p}_i$ for $i \dots N$ for K values with a progressive level of entropy $\mathbf{h}_i$ for $i \dots N$ has been obtained by modifying a starting reference distribution p_{ref}. A description of the algorithm is presented below 5.5.

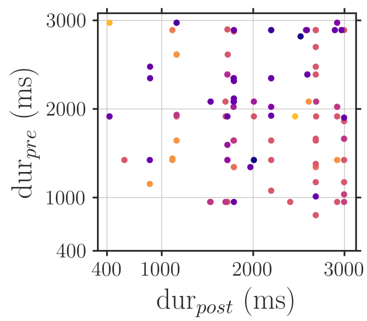

Figure 8: PHASE DURATIONS AND FITNESS - *duration of the first and second phase, over the x and y axis respectively, for each individual of the top-scoring half of the population. The color indicates the fitness, with yellow being the highest.*

Algorithm 2: Reward Probability Distribution Generation

Input: Number of distributions N, dimension K
Output: Set of probability distributions $\mathbf{p}_i$ with decreasing entropy
Define:
  $p_{\text{ref}} \leftarrow \sim \mathcal{N}(0.2, 0.13)^K$; `// reference distribution`
  $p_{\text{ref}} \leftarrow \text{clip}(p_{\text{ref}}, 0, 1)$; `// clipping`
  $\Lambda \leftarrow \{\lambda_i : 1.3^{-i} \mid i \in \text{linspace}(0, 6, N)\}$; `// softmax temperature`
Initial Setup: for $i \leftarrow 0$ *to* N **do**
  $\mathbf{z} \leftarrow p_{\text{ref}}$; `// copy reference distribution`
  $\mathbf{z} \leftarrow \exp(\lambda_i * \mathbf{z}) * \left(\sum_j \exp(\lambda_i * z_j)\right)^{-1}$; `// softmax`
  $\mathbf{z} \leftarrow \mathbf{z} * \sum_j z_j$; `// rescaling`
  $\mathbf{z} \leftarrow \text{clip}(\mathbf{z}, 0, 1)$; `// clipping`
end
return $\mathbf{p}_i$

Given the reference p_{ref} and its entropy H_{ref}, it is possible to quantify the entropy difference with respect to each derived distribution i as $\Delta H = H_{\text{ref}} - H_i$. This measure is informative of how much the best arm stands out from the others.

In Figure 9 are illustrated examples of distributions with decreasing levels of entropy difference with the reference through an heatmap in 9**a**. Both plots highlight the blending of the probability associated with the best arm with the other ones, eventually becoming indistinguishable from the rest. In the MAB taks, this increase in entropy translates in a higher difficulty in picking the best option, although the other side of the medal is that regret also decreases, since the cost of selecting a non-optimal arm is less impactful on performance. Plot 9**b** shows the relationship between the scaling of the difference between the best arm and the rest.

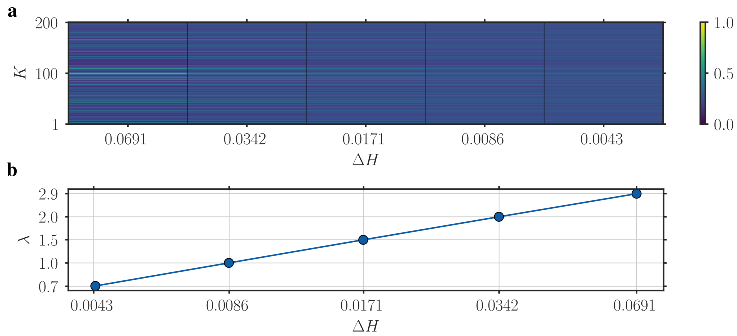

Figure 9: Showcase of distribution entropy - **a**: *visualization of each distribution as heatmaps (columns), the entropy differences (x-axis) and arm indexes (y-axis); note that the entropy of distribution i is inversely proportional to the difference ΔH* - **b**: *log-log plot of the entropy differences (x-axis) against the scaling λ of the top-arm with respect to the others (y-axis).*

5.6 Weight update dynamics

We also analyzed the weight update dynamics of the model over the rounds. In figure 10, we plotted the evolution of the total weight ΔW^{UV} over time, averaged over 20 simulations, and smoothed over 30 rounds. The results show that the model can quickly adapt to new reward distributions. It is also able to maintain the optimal policy over time, with the weights remaining approximately stable. The quantity of updates ΔW_k^{UV}, which in each round is applied to one connection k, changes sign according to the collected reward, with its magnitude higher at the beginning of the trials. Initially, the sign is mostly positive (potentiation) since the weights start at zero, and after some uncertainty a consistently preferred arm emerges. However, when the reward distribution switches, a regular series of suboptimal choices with respect to the new distribution is made, leading to zero reward. This causes an accumulation of negative sign weight updates (depression), eventually causing the value of the preferred arm to drop. In the meantime, other options are probed until another sequence of choices converges to another arm, promoted by a trail of positive weight updates.

This behaviour is consistent with the low entropy levels observed in the previous analysis.

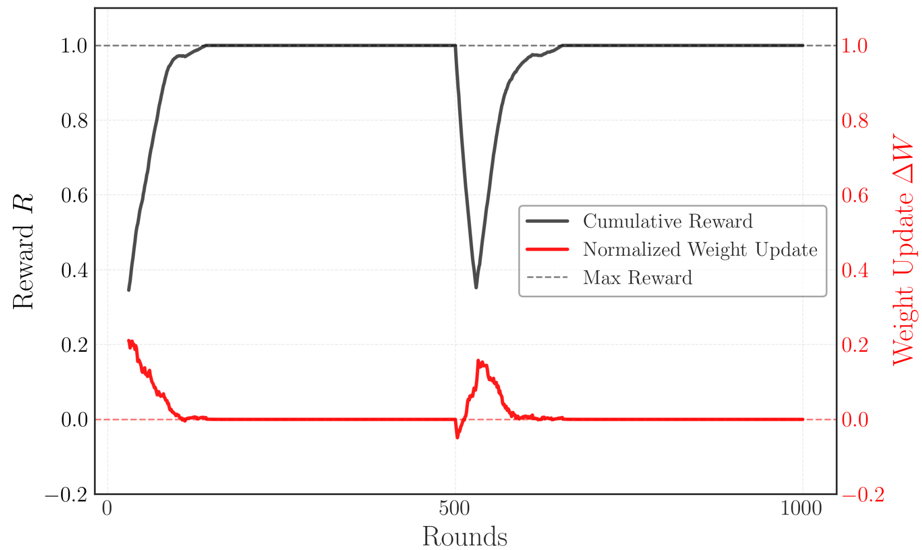

Figure 10: Weight update developement for the model *The plot displays the weight update quantity ΔW_k^{UV} for each round (blue line), smoothed as a 20-steps moving average. It is also reported the average reward in a window of 30 rounds (orange line). The results have been obtained averaging over 30 iterations.*